



OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

Semester 2 Examinations 2025/2026

Course Instance Codes	3BA1, 4BDA1, 4BME1, 1OA2, 1OA3, 4BCT1, 4BMS2, 4BS2
Exams	3rd and 4th year (Arts and Science), Study Abroad / International Exchange
Module Code	CS402
Module	Cryptography
External Examiner	Dr Michael Mackey
Internal Examiners	Dr J. Cruickshank *Dr A. Carnevale
<u>Instructions:</u>	Answer all four questions. Each question carries 25 marks. All workings must be shown.
Duration	2 hours
No. of Pages	4 pages, including this one
Discipline	Mathematics
<u>Requirements:</u>	No special requirements
Release to Library	Yes
Release in Exam Venue	Yes
MCQ	No
Statistical / Log Tables	No
Non-programmable Calculator	Yes
Mathematical Tables	No
Graph paper	No

Q1. The following table may be useful throughout this question. Note that the final character $_$ indicates a space.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	$_$
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

- (a) The following ciphertext string was produced using a shift cipher over the 27-letter alphabet represented in the table above.

O_NNMXXVIRAIBQNIO_NNMXXVIBXIAJGIBQJBIBEXIYUCAIBEXIVJTNIOXC

By conducting a frequency analysis of a larger sample of ciphertexts, it has been found that the plaintext symbol **O** corresponds to the ciphertext symbol **X**. Determine the **first word** of the plaintext string corresponding to the ciphertext string given above. [7 marks]

- (b) Identify $\mathbb{Z}_{27} = \{0, \dots, 25, 26\}$ with $\{A, \dots, Z, _ \}$ as in the above table. The ciphertext **AZMZ** was obtained using a 2-dimensional Hill cipher over $\mathbb{Z}_{27}$ with key

$$\begin{bmatrix} 5 & 7 \\ 4 & 7 \end{bmatrix}.$$

Find the corresponding plaintext. [9 marks]

- (c) Can the matrix

$$\begin{bmatrix} 4 & 7 \\ 2 & 17 \end{bmatrix}$$

be used as the key of a 2-dimensional Hill cipher over $\mathbb{Z}_{27}$? Justify your answer. [4 marks]

- (d) An affine cipher (of dimension 1) over $\mathbb{Z}_{150}$ is used to encrypt messages. Find the number of possible keys. [5 marks]

-
- Q2.** (a) i. Briefly describe what a *perfectly secure* (symmetric) cryptosystem is. [3 marks]
- ii. State (without proof) *Shannon's Theorem* on perfectly secure cryptosystems. [4 marks]
- iii. Do perfectly secure cryptosystems exist? Provide a brief justification of your answer. [3 marks]
- (b) i. Explain what is meant by a *linear feedback shift register (LFSR)* of length L . [3 marks]
- ii. Define what is meant by the *connection polynomial* of an LFSR. [3 marks]
- (c) A linear feedback shift register of length 3 with connection polynomial

$$C(X) = 1 + X^2 + X^3$$

over $\mathbb{Z}_2$ is used to produce a pseudo-random sequence of binary digits with initial values $s_0 = 1$, $s_1 = 0$, and $s_2 = 1$. Determine the least period of the resulting infinite sequence $s_0, s_1, \dots$. Show your work. [9 marks]

- Q3.** (a) Describe the key generation, encryption, and decryption steps of the *RSA* cryptosystem. [5 marks]
- (b) For each of the following pairs (n, e) , decide (with justification) whether it is a valid RSA public key:
- i. $(n, e) = (187, 7)$.
- ii. $(n, e) = (289, 15)$.
- iii. $(n, e) = (143, 3)$. [6 marks]
- (c) Alice encrypts the message $m = 3$ using the RSA cryptosystem and Bob's public key $(n, e) = (187, 7)$. Find the corresponding ciphertext (as an element of $\mathbb{Z}_{187}$). [4 marks]
- (d) Briefly explain the meaning of the following statement: "*The existence of an efficient algorithm for integer factorisation would render the RSA cryptosystem insecure.*" [5 marks]
- (e) Describe the *Fermat primality test*. [5 marks]

-
- Q4.** (a) Define what it means for a group to be *cyclic*. State (without proof) Gauss's characterisation of those $n \geq 1$ for which the group $\mathbb{Z}_n^\times$ is cyclic. [5 marks]
- (b) Alice and Bob use the cyclic group $\mathbb{Z}_{121}^\times$ and its generator $g = 8$ to perform a *Diffie-Hellman key exchange*. Alice secretly chooses $a = 3$ and Bob secretly chooses $b = 5$. Determine their shared key (as an element of $\mathbb{Z}_{121}$). [5 marks]
- (c) Describe the key generation, encryption, and decryption steps of the *ElGamal* cryptosystem. [5 marks]
- (d) Let E be the set of points of an elliptic curve over $\mathbb{R}$ defined by the equation

$$y^2 = x^3 + ax + b.$$

- i. Describe (either geometrically or algebraically) the usual operation of addition on E which turns E into an abelian group. [5 marks]
- ii. For $a = -1$ and $b = 0$, find all points $P \in E$ that satisfy $P + P = \mathcal{O}$, where $\mathcal{O}$ denotes the point at infinity. [5 marks]